

PE6064**[6579]-S-372****T.E. (Mechanical) (Mechanical-Sandwich) (Insem)****DESIGN OF MACHINE ELEMENTS****(2019 Pattern) (Semester - I) (302043)****Solution and Scheme of Marking****UNIT-I**

- Q.1 a) Diameter of Rod & Standard Dimension for Knuckle Joint – 01 mark
Failure of the knuckle pin in shear – 01 mark
Failure of the single eye or rod end in tension – 01 Mark
Failure of the single eye or rod end in crushing – 01 Mark
Failure of the single eye or rod end in shearing – 01 Mark
Failure of the Double eye or forked end in tension – 01 Mark
Failure of the Double eye or forked end in crushing – 01 Mark
Failure of the Double eye or forked end in shearing – 01 Mark
b) Define Factor of Safety - 01 Mark
Service factor – 01 Mark
Factors to be considered while selecting the factor of safety – 05 Marks

OR

- Q.2 a) Calculate W, P, RF – 02 Mark
Design of pin at Fulcrum pin Point – 02 Marks
Dimension of Boss – 01 Marks
Design of pin at Effort Point – 01 Marks
Design of pin at Load Point – 02 Marks
Dimension of cross section of lever arm – 02 Marks
b) Dimension of C/S – 03 Marks
Direct shear stress – 01 Marks
Bending Stress – 01 Mark

UNIT-II

- Q.3 a) Shear stress – 01 Mark
Torque transmitted – 01 Mark
Vertical Load on Pulley – 01 Mark
Design of shaft on the basis of strength – 03 Marks
Design of shaft as per torsional rigidity – 01 Mark
Design of shaft as per Lateral rigidity – 01 Mark
b) Explain function of the key – 01 Mark
What is the effect of keyways cut into the shaft – 02 Marks
Explain the different types of Keys with neat diagrams - 04 Marks

OR

- Q.4 a) Key dimensions – 02 Marks
Hub length – 02 Marks
Thickness of flange – 02 Marks
Diameter of bolts – 02 Marks
Power transmitted – 02 Marks
b) Derivation for $\sigma_c = 2\tau$ - 05 Marks

